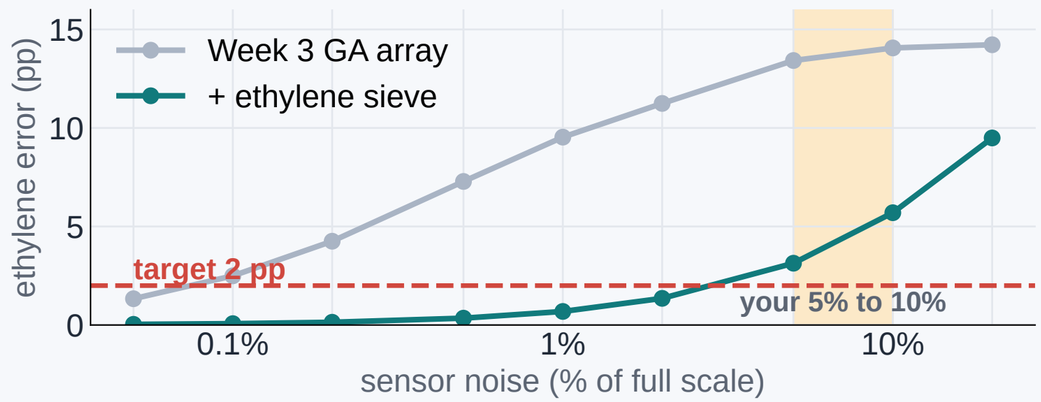
KEN MA / DR. A. J. MEDFORD / OCTOBER 11, 2026

Week 4: an ethylene-selective sensor fixes ethylene, not ethane

Your question or idea	Answer
Lit search: ethylene-selective MOFs	Strongest found: UTSA-280 sieving (apparent selectivity over 10,000) and Ag(I) π-complexation (PAF-1-SO3Ag, 27 to 125). D-MOPH has no metal-site binding.
Best known material in the array	As a 13th sensor, the sieve cuts ethylene error from 9.53 to 0.69 pp at 1% noise (fresh set: 0.73). Metal sites on the best host: 0.71 pp at selectivity 27, 0.94 at 5.
Noise as a sensitivity variable	Without the sieve, the three lightest gases break down below 1% noise. With it: ethylene 10.5%, methane and ethane about 2%, propane and propylene about 5%, heavier gases 8 to 17%.
Pruning toward 4-6 MOFs	8 gases need at least 7 sensors. 6 MOFs plus the sieve keep ethylene at 0.69 pp and overall error at 1.91 pp (all 13: 1.69).

Ethylene error vs sensor noise (pp)



Why ethane is still the limit

Each MOF's full scale is set by the heavy gases, so methane and ethane signals sit near the noise. With the sieve, ethane error is still 3.99 pp and the ratio error 39% (target 10%).

Caveats: literature selectivities come from isotherms, not dilute-limit Henry ratios. The gain depends on the host (median of 464 hosts: 8.89 pp). A sieve that also took up methane at 30% of its ethylene response still gives 1.36 pp.

Next week: lock in the 7-sensor array around the ethylene sensor.

Held-out mixtures; hosts and pruning chosen on development mixtures only. Breakdown: error reaches half of always guessing the average composition. Sources: Lin et al., Nat. Mater. (UTSA-280); Li et al., JACS (PAF-1-SO3Ag); Day and Wilmer, Sensors (GA arrays, 5% noise assumed).